The CTMT Morphism State:
Transport Monodromy and a Family of Diagnostics

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Abstract

We report the current morphism state of the Chronotopic Metric Theory (CTMT) transport program after testing resolved loop holonomy, fixed-sector persistence, winding health, self-reference stability, null-curvature diagnostics, and quotient-style Čencov stability on synthetic coil and Navier-style comparison batteries.

The central conclusion is deliberately conservative. The experiments do not support a single scalar invariant that finalizes the CTMT morphism class. Instead, they support a structural interpretation: admissible transport morphisms preserve a family of diagnostics evaluated on a common loop operator. The primary computable object is the transport monodromy

$$H_\gamma = T_n \cdots T_2 T_1,$$

associated with a closed transport loop. The observed diagnostics are different functionals of this object or of structures transported with it: operator defect, fixed-sector dimension, winding, self-reference/commutator compatibility, null-curvature stability, and quotient-style Čencov stability.

The coil battery shows a clear hierarchy. Radius scaling preserves fixed-sector persistence, winding, and self-reference, but fails operator loop holonomy and null/quotient structural criteria. Thus fixed-sector survival is not transport identity. The Navier-style comparison shows that resolved operator holonomy is more portable than coil-calibrated protected-sector overlap. This identifies the missing mathematical object more sharply: CTMT currently has computable transport monodromy and diagnostic functionals, but not yet a canonical transport bundle, connection, or system-independent protected sector.

Therefore the present result is a tested-class morphism statement, not a universal theorem. CTMT morphism admissibility is currently best described as preservation of a family of transport diagnostics, with the possibility that a future higher-order transport object explains these diagnostics as projections.

Contents

1 Purpose and scope

The purpose of this note is to report the current morphism state of CTMT after the holonomy and fixed-sector stress tests. It is not a proof that CTMT morphisms have been universally derived.

The guiding question is:

> Are CTMT morphisms characterized by one invariant, or by preservation of a family of functionals associated with closed transport loops?

The result is the second. The experiments strongly disfavor the hypothesis that a single scalar invariant finalizes morphism admissibility. They instead support the view that transport identity is structural: different diagnostics detect different failure modes.

Remark 1 (Terminology)**.** We use the term *transport monodromy* for the computable closed-loop operator

$$H_\gamma = T_n \cdots T_1.$$

This is weaker and safer than claiming a fully developed bundle holonomy theory. CTMT currently has transport maps and closed-loop composition. It does not yet have a formally defined bundle, connection, or curvature tensor.

2 Audit background and relevant gaps

The CTMT audit list identifies unresolved structures that are directly relevant here:

- structural transport signature completeness is not proven;
- quotient-style Čencov stability is not shown to define a general equivalence relation;
- null-curvature diagnostics are only operational proxies;
- resolved/null decompositions are not proven stable under arbitrary morphisms;
- window-atlas consistency is not proven;
- self-reference stability is operational, not yet derived from a fixed-point theorem;
- Fisher geometry is local whereas transport is not.

The tests in this note should be understood as an attempt to locate the missing structure behind those diagnostics. They do not close all gaps.

3 Primary computable object: transport monodromy

Let a, b, c be local transport charts, operating windows, or regimes. Let J_a, J_b, J_c denote local Jacobians of the forward map in those charts.

Definition 1 (Resolved transition map)**.** For two charts a, b, define the regularized resolved transition map

$$T_{ab} = \arg\min_{T} \left(\|J_b T - J_a\|_{\mathrm{F}}^2 + \lambda \|T\|_{\mathrm{F}}^2 \right). \tag{3.1}$$

Definition 2 (Transport monodromy)**.** For a closed loop

$$\gamma : a \to b \to c \to a,$$

define the transport monodromy

$$H_\gamma = T_{ca} T_{bc} T_{ab}. \tag{3.2}$$

The raw operator H_γ is representation-dependent. If the local representation is changed by P, then a loop operator may transform as

$$H'_\gamma = P^{-1} H_\gamma P.$$

Thus the matrix itself is not the invariant. Natural diagnostics include operator norms, spectral data, determinant, trace, fixed subspaces, and compatibility with additional structures.

Remark 2 (Why not claim a connection)**.** A connection differentiates or transports sections in a defined bundle. Equation (3.1) gives computable transition operators between local Jacobian charts. This is enough to define closed-loop monodromy, but not enough to claim a canonical geometric connection.

4 Transport observables

Definition 3 (Transport observable)**.** Let V be a finite-dimensional transport representation space and let $H_\gamma \in \mathrm{End}(V)$ be a transport monodromy. A transport observable is a functional

$$\Phi : \mathrm{End}(V) \longrightarrow Y, \tag{4.1}$$

where Y is a target space such as $\mathbb{R}$, a spectrum, a subspace Grassmannian, or a Boolean gate.

The diagnostics in the present CTMT battery are interpreted as transport observables or as structure-dependent observables associated with H_γ:

Diagnostic	Functional form	Detects
Operator loop defect	$\Phi_H(H) = \|H - I\|$	accumulated transport defect
Fixed sector	$\Phi_F(H) = \ker(H - I)$	protected subspace persistence
Spectral near-identity count	$\#\{\lambda_i : \lvert\lambda_i - 1\rvert < \epsilon\}$	near-return modes
Determinant/log-volume	$\lvert \log \lvert \det H \rvert \rvert$	volume distortion
Winding	phase accumulation along loop	topological transport health
Self-reference	smoothing/closure compatibility	recursive consistency
Null-curvature	null-sector contamination test	hidden structural drift
Quotient-Čencov	class variance under reductions	quotient stability

Definition 4 (Transport-separating family)**.** A finite family of transport observables

$$\mathcal{I} = \{\Phi_1, \ldots, \Phi_k\}$$

is transport-separating on a tested class $\mathcal{X}$ if for every pair of tested transport classes that must be distinguished, at least one Φ_i distinguishes them within the stated tolerance.

Remark 3. This definition is intentionally finite and tested-class. It does not require completeness over all possible morphisms or all possible forward models.

5 Diagnostics used in the present battery

5.1 Operator loop defect

The operator defect is

$$D_H(\gamma) = \|H_\gamma - I\|_{\mathrm{F}}. \tag{5.1}$$

This diagnostic directly measures accumulated closed-loop transport defect.

5.2 Fixed-sector persistence

The fixed sector is

$$\mathcal{F}_\gamma = \ker(H_\gamma - I). \tag{5.2}$$

Numerically, the approximate fixed dimension is estimated by

$$d_{\mathrm{fix}}(\gamma) = \#\{\sigma_i(H_\gamma - I) \le \epsilon_{\mathrm{fix}}\}, \tag{5.3}$$

and the eigenvalue near-one count by

$$n_1(\gamma) = \#\{\lambda_i(H_\gamma) : |\lambda_i - 1| \le \epsilon_\lambda\}. \tag{5.4}$$

5.3 Protected-sector overlap

If a protected sector projector P_{prot} is available and Q_{fix} is an orthonormal basis for the approximate fixed sector, define

$$O_{\text{prot}}(\gamma) = \operatorname{tr}(Q_{\text{fix}}^{\top} P_{\text{prot}} Q_{\text{fix}}). \tag{5.5}$$

The Navier-style tests show that this quantity is not yet portable across systems when the protected sector is calibrated from the coil model.

5.4 Winding health

For a complex transport series z_k, define

$$W(z) = \frac{\operatorname{unwrap}\arg(z_N) - \operatorname{unwrap}\arg(z_0)}{2\pi}. \tag{5.6}$$

Winding health requires $|W - 1| \leq \varepsilon_W$ and non-collapsed amplitude.

5.5 Self-reference health

The self-reference transform is

$$z \mapsto z + \alpha\big(\operatorname{smooth}(z) - z\big). \tag{5.7}$$

Self-reference health requires a reduction of loop inconsistency after this transform.

A more structural future version should test a commutator defect

$$D_S(H) = \|SH - HS\|, \tag{5.8}$$

where S is a well-defined smoothing or recursive-closure operator acting on the same representation as H. In the present battery, the self-reference test remains operational rather than a proven commutator invariant.

6 Synthetic coil battery

The coil battery includes six scenarios:

Scenario	Intended class
baseline	Gauge candidate
current scaling	Gauge candidate
radius scaling	Dangerous compensated
compensated attack	Dangerous compensated
causal violation	Violation
unit leak	Violation

Earlier CTMT coil diagnostics showed that radius scaling and compensated attack can preserve Fisher monotonicity, winding, and self-reference while failing null-curvature and quotient-style Čencov stability. This motivated the loop and fixed-sector tests.

7 Coil results: hierarchy of diagnostics

Empirical finding 1 (Fixed-sector persistence is not transport identity)**.** On the coil battery, radius scaling preserves the approximate fixed sector, winding, and self-reference health, but fails operator loop holonomy and previously failed null-curvature/quotient-style Čencov diagnostics. Therefore fixed-sector persistence is not sufficient for CTMT transport identity.

Scenario	Operator holonomy	Fixed sector	Winding	Self-ref	Outcome
baseline	PASS	PASS	PASS	PASS	accept
current scaling	PASS	PASS	PASS	PASS	accept
radius scaling	FAIL	PASS	PASS	PASS	refuse
compensated attack	FAIL	FAIL	PASS	PASS	refuse
causal violation	FAIL	FAIL	FAIL	FAIL	refuse
unit leak	FAIL	PASS	FAIL	FAIL	refuse

Table 1: Coil battery hierarchy. Radius scaling is decisive: fixed sector, winding, and self-reference survive, but operator holonomy fails.

Scenario	D_H	d_{fix}	n_1	O_{prot}	W	Self-ref err
baseline	2.855×10^{-2}	3	3	3.0	0.9825	0.0153
current scaling	3.115×10^{-2}	3	3	3.0	0.9825	0.0215
radius scaling	6.024×10^{-2}	3	3	3.0	0.9819	0.0185
compensated attack	1.509×10^{-1}	2	2	2.0	0.9817	0.0201
causal violation	8.043×10^{3}	2	1	2.0	1.3021	1.5817
unit leak	5.359×10^{-2}	3	3	3.0	1.0692	0.6386

Table 2: Numerical coil diagnostics. Operator holonomy separates radius scaling from gauge candidates even when fixed-sector persistence and topology survive.

8 Navier-style comparison

The Navier-style suite is synthetic. It is designed to test whether a candidate invariant wrongly rejects structured seepage or node-preserving transport. The tested scenarios are:

Scenario	Intended class
laminar gauge	Gauge candidate
mild seepage	Seepage
Kolmogorov rank thinning	Seepage
node preserving	Node preserving
shock violation	Violation

Resolved operator holonomy remains stable for laminar, seepage, rank-thinning, and node-preserving cases, while shock violation fails. However, a coil-calibrated protected-sector overlap rejects all Navier-style cases. This demonstrates that the protected sector is not yet defined in a system-independent way.

Navier-style case	Operator loop	Coil-calibrated fixed-sector gate	Interpretation
laminar gauge	PASS	FAIL	protected sector not portable
mild seepage	PASS	FAIL	structured seepage not rejected by operator loop
Kolmogorov rank thinning	PASS	FAIL	rank thinning remains loop-stable
node preserving	PASS	FAIL	node-like transport remains loop-stable
shock violation	FAIL	FAIL	hard transport defect

Table 3: Navier-style comparison. Operator loop holonomy is portable in this synthetic test; protected fixed-sector overlap is not.

Remark 4 (Meaning of the Navier-style result)**.** The Navier-style result does not invalidate fixed-sector logic. It shows that *which sector is protected* is representation- and system-dependent unless CTMT defines a system-independent protected transport object.

9 Relation to null-curvature and quotient-style Čencov stability

The previous CTMT coil battery used the structural signature

$$\Sigma_{\mathrm{tr}} = (1_N, 1_W, 1_S, 1_C), \tag{9.1}$$

where the entries correspond to null-curvature stability, winding, self-reference, and quotient-style Čencov stability.

The prior pattern was:

Scenario	Fisher	Null	Winding	Self-ref	Čencov
baseline	PASS	PASS	PASS	PASS	PASS
current scaling	PASS	PASS	PASS	PASS	PASS
radius scaling	PASS	FAIL	PASS	PASS	FAIL
compensated attack	PASS	FAIL	PASS	PASS	FAIL
causal violation	FAIL	FAIL	FAIL	FAIL	FAIL
unit leak	FAIL	FAIL	FAIL	FAIL	FAIL

The new monodromy diagnostics do not replace this structure. They explain why this structure is a family. Operator holonomy detects loop-level transport defect. Fixed sector detects invariant subspace persistence. Winding detects topological health. Self-reference detects recursive closure. Null-curvature and quotient-style Čencov stability detect structural transport-class failures not visible to Fisher monotonicity alone.

Remark 5 (Čencov caution)**.** Čencov's theorem is not inherited directly. CTMT morphisms are not yet proven Markov morphisms, and CTMT quotient equivalence is not yet proven to satisfy the required axioms. The current use is quotient-style and operational.

10 Morphism as preservation of a transport-observable family

The present evidence supports the following tested-class definition.

Definition 5 (CTMT transport-observable family)**.** On a tested class $\mathcal{X}$, let

$$\mathcal{I}_{\mathrm{CTMT}} = \{\Phi_H, \Phi_F, \Phi_W, \Phi_S, \Phi_N, \Phi_C\}, \tag{10.1}$$

where

- Φ_H is operator loop holonomy defect;
- Φ_F is fixed-sector persistence;
- Φ_W is winding health;
- Φ_S is self-reference / recursive closure health;
- Φ_N is null-curvature stability;
- Φ_C is quotient-style Čencov stability.

Definition 6 (Tested-class CTMT morphism)**.** A morphism f is CTMT-admissible on $\mathcal{X}$ if it preserves all diagnostics in the tested transport-observable family within tolerance:

$$f \in \mathrm{Mor}_{\mathrm{CTMT}}(\mathcal{X}) \quad \Longleftrightarrow \quad f^*\Phi = \Phi \quad \text{for every } \Phi \in \mathcal{I}_{\mathrm{CTMT}}. \tag{10.2}$$

This definition is intentionally not scalar. It says the morphism is the structure-preserving map class, not an invariant itself.

11 Tested-class theorem

Theorem 1 (Loop-refined tested-class morphism separation)**.** *On the present synthetic coil battery, the family*

$$\mathcal{I}_{\mathrm{CTMT}} = \{\Phi_H, \Phi_F, \Phi_W, \Phi_S, \Phi_N, \Phi_C\}$$

accepts baseline and current scaling, and refuses radius scaling, compensated attack, causal violation, and unit leak.

Proof 7. The proof is by finite evaluation on the tested coil suite.

Baseline and current scaling pass operator loop holonomy, fixed-sector persistence, winding, self-reference, null-curvature, and quotient-style Čencov stability.

Radius scaling passes fixed-sector persistence, winding, and self-reference, but fails operator loop holonomy and previously fails null-curvature and quotient-style Čencov stability. Compensated attack fails operator loop holonomy and fixed-sector persistence and also fails null-curvature and quotient-style Čencov stability. Causal violation fails operator holonomy, fixed-sector, winding, and self-reference. Unit leak fails operator holonomy and transport health, and previously fails all structural indicators.

Therefore each accepted scenario is a gauge candidate and each dangerous or violation scenario is refused by at least one necessary diagnostic in the family.

Remark 6 (No universality claim)**.** The theorem is a tested-class statement. It does not prove that $\mathcal{I}_{\mathrm{CTMT}}$ is minimal, complete, or sufficient outside the tested coil suite.

12 What the experiments falsified

The experiments falsified several stronger hypotheses.

12.1 Single-scalar finalization

No tested scalar diagnostic alone finalizes the morphism class. Operator defect, fixed-sector dimension, winding, and self-reference detect different failure modes.

12.2 Fixed-sector sufficiency

Fixed-sector persistence is not sufficient. Radius scaling preserves the fixed sector but fails operator holonomy and class-level structural diagnostics.

12.3 Pure metric holonomy

Fisher-metric whitening holonomy is too flat and too canonical to be a decisive transport-class invariant.

12.4 Portable protected-sector overlap

Protected fixed-sector overlap calibrated on the coil model does not transfer to the Navier-style suite. A system-independent protected transport object remains missing.

13 Suggested upper structure

The evidence suggests, but does not prove, the existence of a higher-order transport object whose projections yield the observed diagnostics.

The conservative formulation is:

CTMT currently has a computable transport monodromy H_γ and a family of transport observables $\Phi_i(H_\gamma)$. The observed hierarchy suggests that these observables may be projections of a common higher-order transport structure, but that structure is not yet formally derived.

A possible future object would have to explain why the following appear as coordinated but non-equivalent diagnostics:

Projection / functional	Current diagnostic
operator norm or spectral radius	loop defect
kernel / near-one eigenspace	fixed sector
phase / argument accumulation	winding
commutator with smoothing/closure map	self-reference
null restriction	null-curvature stability
quotient variation under reductions	quotient-style Čencov stability

Remark 7 (Future geometric language)**.** It may eventually be natural to describe this in terms of a transport bundle, connection, or curvature. The present paper does not claim those structures. It only establishes the computable lower-level fact: closed-loop transport monodromy has multiple diagnostics, and the diagnostics form a tested-class separating family.

14 Non-claims

This paper does not claim:

- that CTMT morphisms are universally derived;
- that a transport bundle or connection has already been constructed;
- that Čencov's theorem directly applies to CTMT morphisms;
- that any single scalar invariant finalizes the morphism class;
- that fixed-sector persistence is sufficient;
- that the invariant family is minimal or complete;
- that the Navier-style synthetic suite is laboratory evidence;
- that the protected sector has a system-independent definition.

15 Falsifiability and next work

15.1 Downward falsification

The tested-class theorem fails if a future coil-like or laboratory suite produces a demonstrably non-gauge scenario that passes the full diagnostic family.

It must also be weakened if a physically gauge-equivalent scenario systematically fails one of the required diagnostics.

15.2 Upward sharpening

The result would be strengthened by constructing a single higher-order transport object $\mathcal{K}_\gamma$ such that

$$\Phi_H, \Phi_F, \Phi_W, \Phi_S, \Phi_N, \Phi_C$$

are all natural projections or functionals of $\mathcal{K}_\gamma$.

15.3 Most important missing definition

The most important missing definition is a system-independent protected transport sector. Without it, fixed-sector overlap remains battery-dependent.

16 Conclusion

The monodromy and holonomy stress tests did not fail the CTMT morphism program. They falsified a stronger and less plausible hypothesis: that one scalar invariant would finalize the CTMT morphism class.

The current evidence supports a more mature structure:

> Admissible CTMT morphisms preserve a family of transport observables evaluated on closed-loop transport monodromy and associated transported structures.

Operator holonomy detects accumulated transport defect. Fixed-sector persistence detects invariant subspace survival. Winding detects topological transport health. Self-reference detects recursive closure. Null-curvature and quotient-style Čencov stability detect structural class failure not visible to Fisher monotonicity alone.

Radius scaling is the decisive example: it preserves fixed sector, winding, and self-reference, but fails operator holonomy and null/quotient structure. Therefore fixed-sector persistence is not transport identity.

The current CTMT morphism state is therefore:

$$\boxed{\text{transport identity is structural, not scalar.}}$$

and the current tested-class morphism definition is:

$$\boxed{\mathrm{Mor}_{\mathrm{CTMT}}(\mathcal{X}) = \{f : f^*\Phi = \Phi \text{ for all } \Phi \in \mathcal{I}_{\mathrm{CTMT}}\}.}$$

This is not a universal final theorem. It is a defensible tested-class morphism statement and a precise guide for the next mathematical construction.

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